



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

CÁC HỆ THỐNG PHÂN TÁN VÀ ỨNG DỤNG

Chapter 7: Cloud Computing

Outline

1. Introduction
2. Cloud computing architecture
3. Technologies and Commercial products
4. Challenges

1. Introduction

Introduction

- rapid development of processing and storage technologies and the success of the Internet →computing resources have become cheaper, more powerful and more ubiquitously available than ever before
- → enabled the realization of a new computing model: **cloud computing**
- resources (e.g., CPU and storage) are provided as general utilities that can be leased and released by users through the Internet in an **on-demand fashion**.

Introduction (2)

- the traditional role of service provider is divided into two:
  - *infrastructure providers*
  - *service providers*
- → Google, Amazon and Microsoft strive to provide more powerful, reliable and cost-efficient cloud platforms
- compelling features:
  - *No up-front investment*
  - *Lowering operating cost*
  - *Highly scalable*
  - *Easy access*
  - *Reducing business risks and maintenance expenses*

Overview of Cloud Computing

- Definitions (by The National Institute of Standards and Technology (NIST):
- *Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.*
- Cloud Computing is not a new technology, but rather a new operations model that brings together a set of existing technologies to run business in a different way

Related technologies

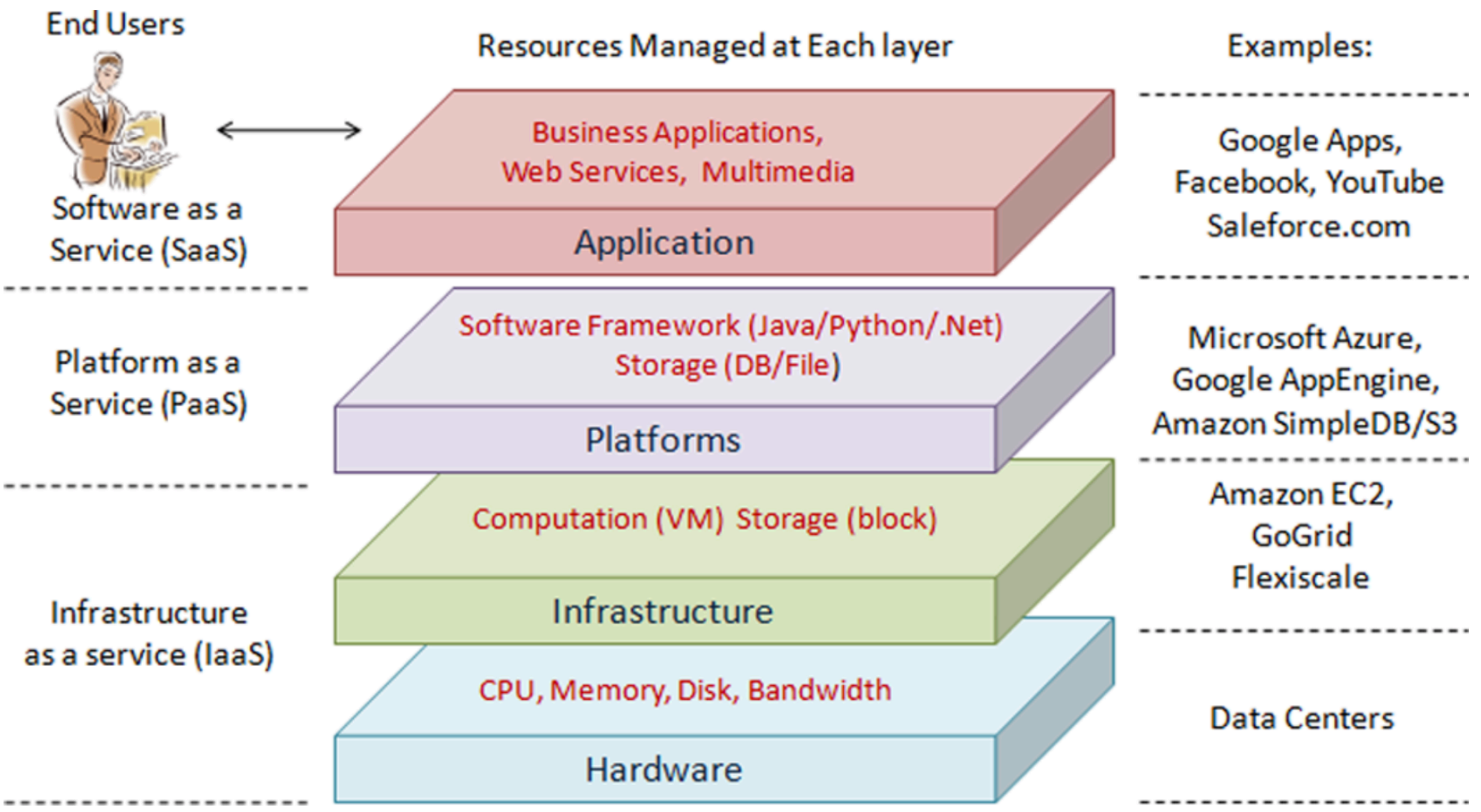
- *Grid Computing*
- *Utility Computing*
- *Virtualization*
- *Autonomic Computing*

Characteristics

- *Multi-tenancy*
- *Shared resource pooling*
- *Geo-distribution and ubiquitous network access*
- *Service oriented*
- *Dynamic resource provisioning*
- *Self-organizing*
- *Utility-based pricing*

2. Cloud computing architecture

2.1. Layered model



Hardware layer

- responsible for managing the **physical resources** of the cloud (physical servers, routers, switches, power and cooling systems, etc.)
- typically implemented in **data centers**
- **A data center**: thousands of servers that are organized in racks and interconnected through switches, routers or other fabrics
- Typical issues:
  - hardware configuration
  - fault- tolerance
  - traffic management
  - power and cooling resource management

Infrastructure layer

- = **virtualization layer**
- Creating a pool of **storage and computing resources** by partitioning the physical resources using **virtualization technologies (**Xen, KVM, and VMware)
- → essential component of cloud computing with many key features (e.g. dynamic resource assignment)

Platform layer

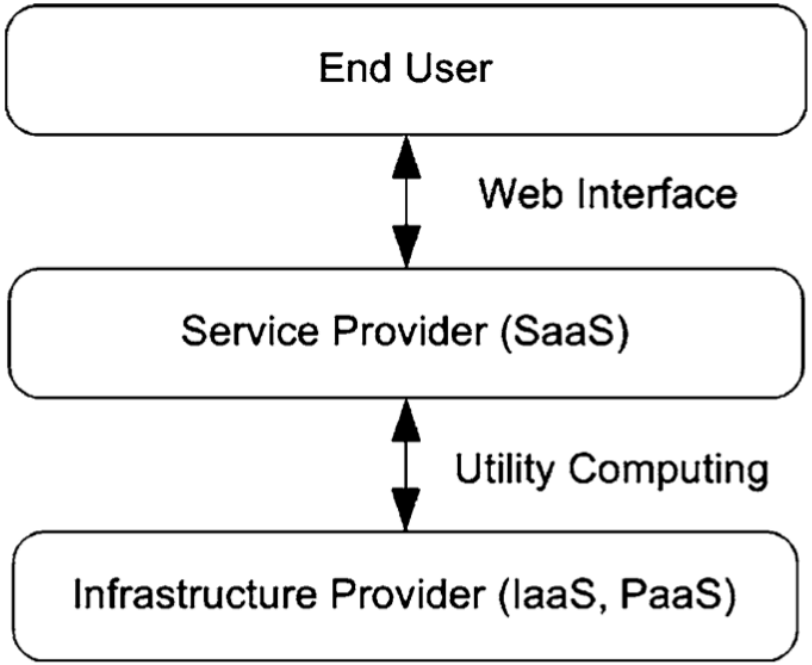
- operating systems and application frameworks
- minimize the burden of deploying applications directly into VM containers
- Example: Google App Engine

Application layer

- Cloud applications
- Different from traditional applications, cloud applications can leverage the **automatic-scaling feature** to achieve better performance, availability and lower operating cost.
- More modular than traditional service hosting environments (dedicated server farms)
- Each layer is loosely coupled with other layers
- Architectural modularity → support a wide range of application requirements while reducing management and maintenance overhead.

2.2. Business model

- every layer can be implemented as a service to the layer above → every layer can be perceived as a customer of the layer below
- 3 categories:
  - Software as a Service (SaaS): Amazon EC2, GoGrid, Flexiscale
  - Platform as a Service (PaaS): Google App Engine, Microsoft Windows Azure, and Force.com
  - Infrastructure as a Service (IaaS): Salesforce.com, Rackspace, and SAP Business ByDesign

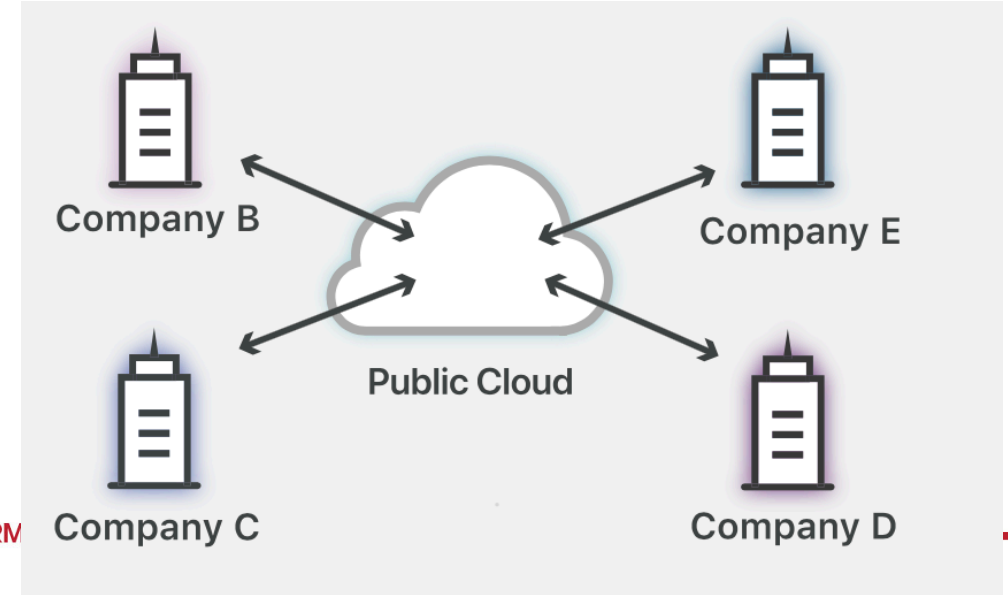


2.3. Types of clouds

- many issues to consider when moving an enterprise application to the cloud environment
  - lowering operation cost
  - high reliability
  - security
- →different types of clouds:
  - *Public clouds*
  - *Private clouds*
  - *Hybrid clouds*
  - *Virtual Private Cloud*

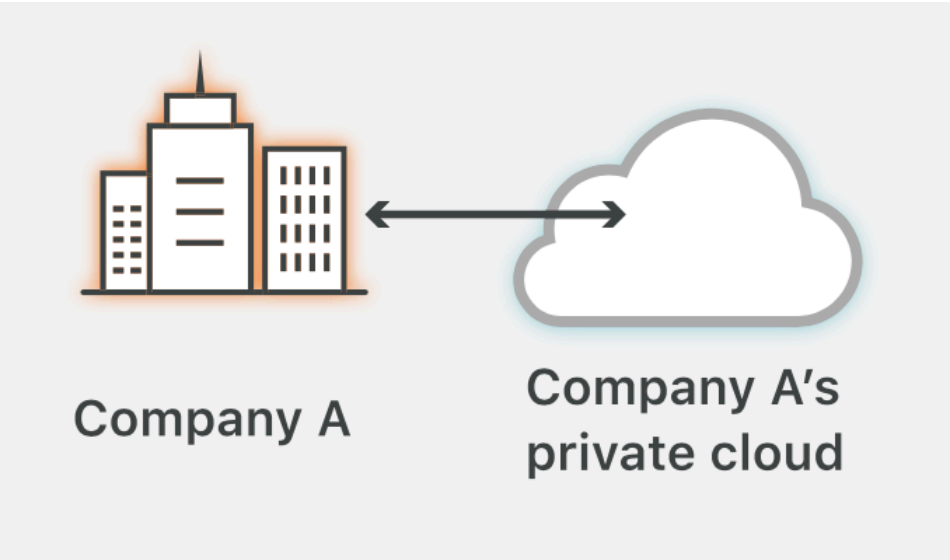
Public clouds

- service providers offer their resources as services to the general public
- key benefits
  - no initial capital investment on infrastructure
  - shifting of risks to infrastructure providers
- Problems:
  - lack fine-grained control over data, network and security settings



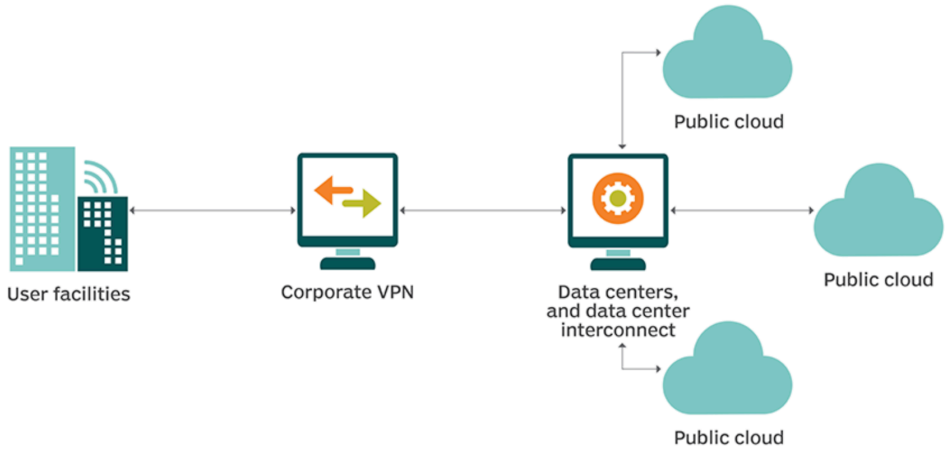
Private clouds

- = internal clouds
- designed for exclusive use by a single organization
- may be built and managed by the organization or by external providers
- offering the highest degree of control over performance, reliability and security
- Problems:
  - criticized for being similar to traditional proprietary server farms
  - do not provide benefits such as no up-front capital costs



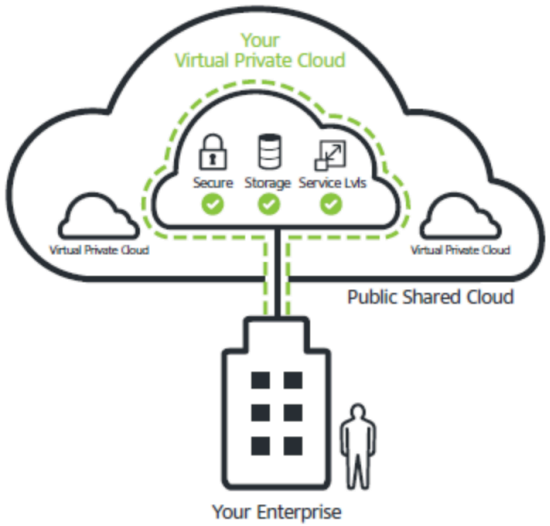
Hybrid clouds

- a combination of public and private cloud models
- part of the service infrastructure runs in private clouds while the remaining part runs in public clouds
- offering more flexibility than both public and private clouds
  - they provide tighter control and security over application data compared to public clouds, while still facilitating on-demand service expansion and contraction
- requiring carefully determining the best split between public and private cloud components



Virtual Private Cloud

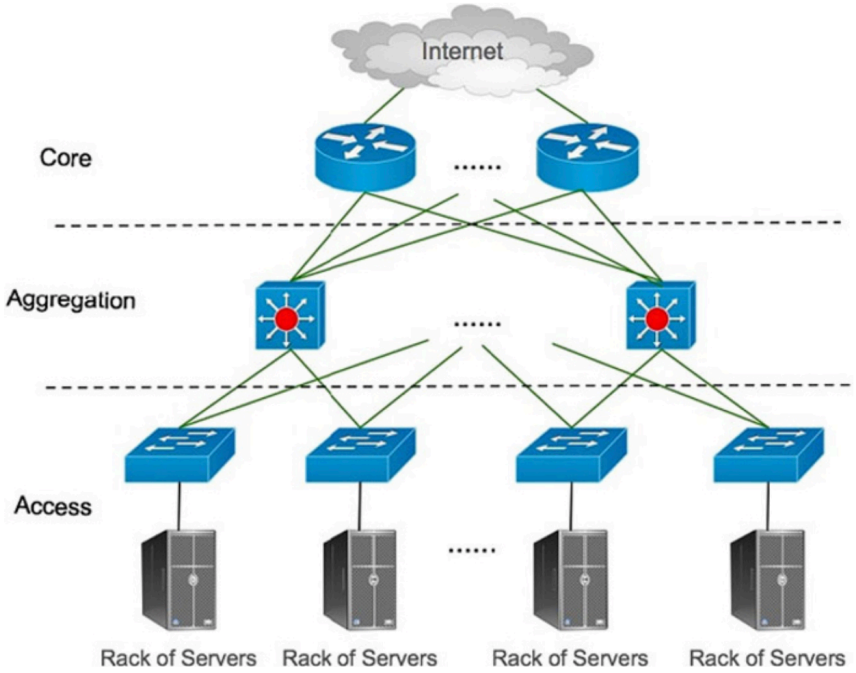
- a platform running on top of public clouds
- leveraging virtual private network (VPN) technology → allowing service providers to design their own topology and security settings (firewall rules)
-



3. Technologies and Commercial products

3.1. Cloud computing technologies (1)

- **Architectural design of data centers**
- Objectives:
  - *Uniform high capacity*
  - *Free VM migration*
  - *Resiliency*
  - *Scalability*
  - *Backward compatibility*
  -



Cloud computing technologies (2)

- **Distributed file system over clouds**
  - Google File System (GFS)
  - Hadoop Distributed File System (HDFS)
- **Distributed application framework over clouds**
  - Google MapReduce
  - Hadoop MapReduce

3.2. Commercial products

- *Amazon EC2*
- *Microsoft Windows Azure platform*
- *Google App Engine*

4. Challenges

4.1. Automated service provisioning (1)

- the capability of acquiring and releasing resources on-demand
- Objective of service providers: allocating and de-allocating resources from the cloud to satisfy its service level objectives (SLOs)
- It is not easy to de- termine how to map SLOs (QoS requirements) to low-level resource requirement (CPU, storage, network)
- the resource provisioning decisions must be made online

Automated service provisioning (2)

- Typical steps for dynamic resource provisioning:
  1. Constructing an application performance model that predicts the number of application instances required to handle demand at each particular level
  2. Periodically predicting future demand and determining resource requirements using the performance model
  3. Automatically allocating resources using the predicted resource requirements
- Various techniques: Queuing theory, Control theory, and Statistical Machine Learning
- Proactive and Reactive Resource Control

4.2. Virtual machine migration

- Virtualization can provide significant benefits in cloud computing by enabling virtual machine migration to balance load across the data center
- process migration techniques
- Xen and VMWare have implemented “live” migration of VMs that involves extremely short downtimes (10ms-1s)
- major benefits of VM migration is to avoid hotspots

4.3. Server consolidation

- effective approach to maximize resource utilization while minimizing energy consumption
- live VM migration technology is often used to consolidate VMs residing on multiple under-utilized servers onto a single server → remaining servers can be set to an energy-saving state

4.4. Energy management

- Designing energy-efficient data centers
- energy-efficient hardware architecture that enables slowing down CPU speeds and turning off partial hardware components
- Energy-aware job scheduling and server consolidation are two other ways to reduce power consumption by turning off unused machines
- A key challenge: achieving a good trade-off between energy savings and application performance

Other challenges

- Traffic management and analysis
- Data security
- Software frameworks
- Storage technologies and data management
- Novel cloud architectures

ĐẠI HỌC BÁCH KHOA

25 YEARS ANNIVERSARY SOICT

VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Questions?



soict.hust.edu.vn/ fb.com/groups/soict